

1 Explainability in Reinforcement Learning

2 Maxime Alaarabiou

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Chapter 1

Experiments

This chapter presents the experimental study of the Clustering Agent Representation (CAR) method introduced in the previous chapter. It focuses on the experimental validation of the method and its ability to consistently identify meaningful conceptual structures. The remainder of this chapter is organized as follows.

- Section 1.1 describes the software implementation of the CAR pipeline.
- Section 1.2 details the experimental configuration, including the environments, the reinforcement learning setup, and the training of the surrogate policies.
- Section 1.3 presents and analyses the results obtained by applying CAR and its metrics.
- Section ?? summarizes the findings and discusses their implications.

1.1 Software Implementation

Deep Learning (DL) and Reinforcement Learning (RL) are inherently empirical fields, in which the effectiveness of a method is assessed through experimentation. For this reason, particular attention is given to the implementation of CAR. It relies on established libraries that provide efficient and widely used implementations of its main components. The implementation is made publicly available as open-source to support the objective and reproducible replication of the reported results.¹

CAR is organized as a pipeline of five stages, shown in Figure 1.1. Each stage is an independent executable, and stages communicate only through artifacts written to disk. This decoupling allows any stage to be rerun or inspected independently. The parameters of each experiment are gathered in a single configuration object, described in Section 1.2. Every result can therefore be traced back to the settings that produced it.

Table 1.1 summarizes the main third-party libraries used at each stage of the pipeline.

¹https://github.com/Maxalaar/Clustering_Agent_Latent_Space

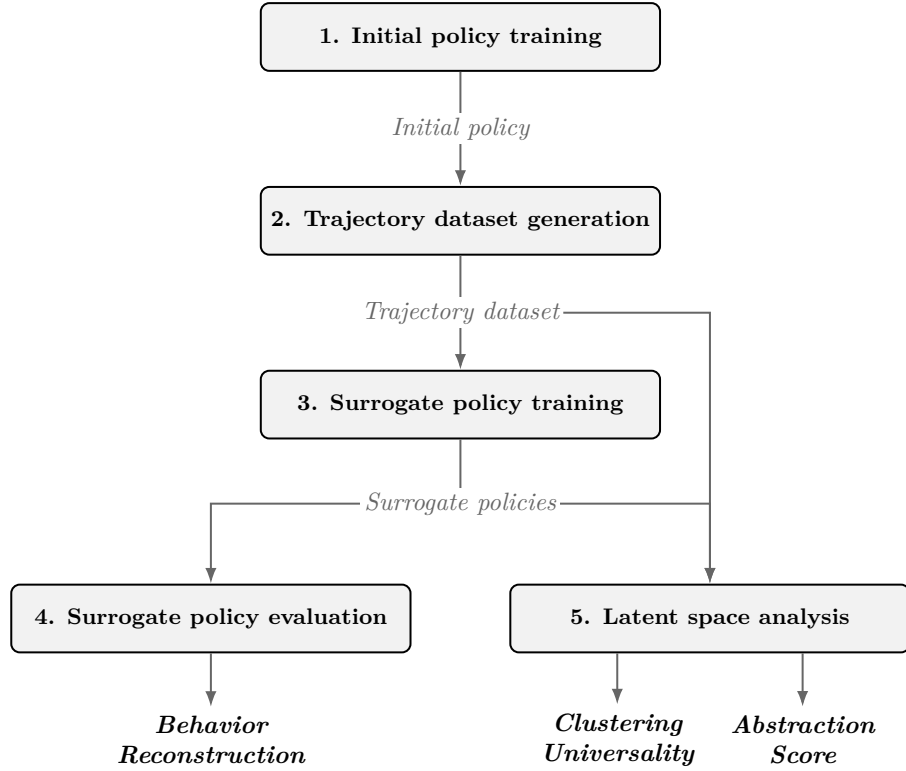


Figure 1.1: The CAR pipeline.

1.1.1 Initial policy training

The first stage trains the initial policy using RLlib [Moritz et al., 2018, Liang et al., 2018]. RLlib provides a highly scalable implementation of RL algorithms and supports a wide range of state-of-the-art methods. It supports distributed training, allowing the computational workload to be spread across multiple processes and machines.

The use of RLlib allows CAR to be applied to any environment implemented using the Gymnasium interface [Towers et al., 2024]. This provides a common interface for environments with different observation and action spaces. This flexibility allows CAR to be applied to a wide range of environments.

By default, the RL algorithm used for the initial policy is Proximal Policy Optimization (PPO) [Schulman et al., 2017]. Other algorithms supported by RLlib, such as Asynchronous Proximal Policy Optimization (APPO) [Espeholt et al., 2018], Deep Q-Networks (DQN) [Mnih et al., 2013], and DreamerV3, can also be selected through the same configuration.

The policy and critic networks are implemented using PyTorch [Paszke et al., 2019]. PyTorch provides efficient GPU computation and supports a wide range of neural network architectures. This provides a standardized implementation that is compatible with the broader DL ecosystem.

The output of this stage is a checkpoint containing the trained policy.

Pipeline Stage	Library	Description
Initial policy training	RLlib [Moritz et al., 2018, Liang et al., 2018]	Distributed RL library used to train the initial policy.
	PyTorch [Paszke et al., 2019]	Deep learning framework implementing the policy and critic networks.
	Gymnasium [Towers et al., 2024]	Common interface used to plug in any RL environment.
Trajectory dataset generation	RLlib [Moritz et al., 2018, Liang et al., 2018]	Distributed environment runners used to collect trajectories in parallel.
	Apache Arrow [Apache Arrow, 2016]	Columnar file format (Parquet) used to store the trajectory datasets.
Surrogate policy training	PyTorch Lightning [Falcon, 2019]	Structured training loop used for the behavior cloning of the surrogate policies.
Surrogate policy evaluation	RLlib [Moritz et al., 2018, Liang et al., 2018]	Wraps each surrogate policy as an RL policy to evaluate it in the environment.
Latent space analysis	cuML [RAPIDS, 2023]	GPU-accelerated clustering algorithms used to analyze the latent representations.

Table 1.1: Main software libraries used at each stage of the CAR pipeline.

1.1.2 Trajectory dataset generation

The second stage builds the dataset used to train the surrogate policies. The distributed execution capabilities of RLlib are used to collect trajectories in parallel. Multiple environment runners execute the initial policy simultaneously, distributing the computational workload across the available resources.

The initial policy is run in the environment, and the interactions between the agent and the environment are recorded. For each step, the dataset stores the observation together with the parameters of the action distribution produced by the policy. Trajectories are stored as Parquet files [Apache Arrow, 2016]. Parquet is a compressed columnar format widely used for large-scale data processing. Its columnar structure allows data to be read efficiently and supports concurrent access from multiple processes. This is particularly useful here because the datasets contain on the order of one million observation–action pairs and cannot be loaded entirely into memory.

1.1.3 Surrogate policy training

The third stage trains the surrogate policies using behavior cloning. This stage therefore converts the original RL problem into a supervised learning problem. The surrogate policies are trained to reproduce the action distributions generated by the initial policy .

The training procedure is implemented using PyTorch Lightning [Falcon, 2019]. PyTorch Lightning provides a structured interface for supervised training while remaining compatible with the PyTorch ecosystem. It handles common training operations such as data loading, optimization, checkpointing, and GPU execution, while allowing the underlying neural network implementation to remain fully controlled.

Once training is complete, the stage produces a set of multiple trained

79 surrogate policies.

80 1.1.4 Surrogate policy evaluation

81 The fourth stage measures how faithfully each surrogate policy reproduces the
 82 behavior of the initial policy. Each surrogate is wrapped so that it can be used
 83 as an RLlib policy and is then evaluated in the original environment. Its mean
 84 cumulative reward over multiple episodes is compared with that of initial policy.
 85 Evaluation episodes are collected using parallel environment runners.

86 This stage provides the value of the *Behavior Reconstruction*. It therefore
 87 verifies whether the surrogate policies adequately reproduce the behavior of the
 88 initial policy before their latent representations are analyzed. If a surrogate does
 89 not reproduce the original behavior sufficiently well, its training can be repeated
 90 before proceeding to the next stage.

91 1.1.5 Latent space analysis

92 The fifth stage performs the main analysis of CAR. The latent representations
 93 produced by the surrogate policies form the point clouds that are analyzed using
 94 clustering algorithms.

95 The clustering step relies on the cuML library [RAPIDS, 2023], which pro-
 96 vides GPU-accelerated implementations of several clustering algorithms. This
 97 significantly reduces the computational cost of clustering and makes it possible
 98 to analyze point clouds containing tens of thousands of representations.

99 The resulting partitions are used to compute the *Clustering Universality* and
 100 the *Abstraction Score*. The outputs of this stage are the resulting metric values
 101 and the material required for the qualitative analysis presented in the following
 102 sections.

103 1.2 Experimental Configuration

104 This section describes the configuration of the experiments reported in this
 105 chapter. It covers the environments, the neural architecture of the policies, and
 106 the main parameters of each pipeline stage. In general, for each environment, a
 107 single initial policy is trained and then imitated by three surrogate policies.

108 1.2.1 Environments

109 All experiments use environments from the Atari benchmark [Bellemare et al., 2013],
 110 accessed through the Arcade Learning Environment. The use of Atari environ-
 111 ments offers several advantages. First, Atari is a widely used benchmark in
 112 the RL community, with many studies using these environments as standard
 113 evaluation settings. This makes the experimental results easier to compare
 114 with existing work. Second, these environments were not specifically designed
 115 for RL, which reduces the risk of introducing biases related to the design of
 116 the environments themselves. Finally, the benchmark provides a diverse set of
 117 well-known environments with varying levels of complexity. This diversity makes
 118 it possible to evaluate CAR across different types of tasks and levels of difficulty.

The configuration of the Atari environments follows common practice in RL research [Machado et al., 2017].

In each environment, the agent receives visual observations in the form of raw pixels. Each image is resized to 64×64 pixels and converted to grayscale to reduce memory requirements. An exception is made for Pac-Man, where color information is important for determining the behavior of other agents in the environment. The original color information is therefore preserved for this environment.

To provide information about the dynamics of the environment, four consecutive observations are stacked along the channel dimension. The resulting observation has a shape of $64 \times 64 \times 4$. This allows the agent to infer the motion of objects from successive observations rather than relying on a single frame.

Frame skipping is also used to reduce the duration of each episode. The agent selects a new action every five frames, while the selected action is repeated for the intervening frames. This provides sufficient temporal resolution for effective control while reducing the number of decisions required during an episode.

The actions correspond to the different buttons available on the Atari controller. The resulting action space is therefore discrete. Since Atari games are fully deterministic, sticky actions are introduced to add stochasticity to the trajectories. With a probability of 0.05, the environment repeats the action selected at the previous step instead of applying the new action. This prevents the agent from simply memorizing a fixed sequence of actions during training.

With the environment configuration established, the next section describes the objective associated with each environment. A visualization of the six environments is provided in Figure 1.2 to give an overview of the experimental settings.

Breakout. The agent moves a paddle at the bottom of the screen (Figure 1.2a) to bounce a ball into a wall of bricks, each broken brick scoring a point, and the ball speeding up as the wall is cleared.

Mario Bros. A single-screen platform game (Figure 1.2b). The player clears each phase by hitting enemies from below to flip them, then kicking them off the platforms before they recover. A central POW block flips every enemy at once but can only be used a few times per phase.

Ms. Pac-Man. The player moves through a maze (Figure 1.2c), eating pellets while avoiding four ghosts, with power pellets briefly making the ghosts edible. The maze structure and the many possible local situations, such as a nearby ghost, an intersection, or an active power pellet, produce a large number of distinguishable states.

Pong. A minimal two-player game (Figure 1.2d) in which the agent controls one paddle and must send the ball past the opponent.

Space Invaders. The agent moves a cannon along the bottom of the screen (Figure 1.2e), firing at descending rows of aliens while dodging their shots and using destructible shields, with a bonus saucer occasionally crossing the top.

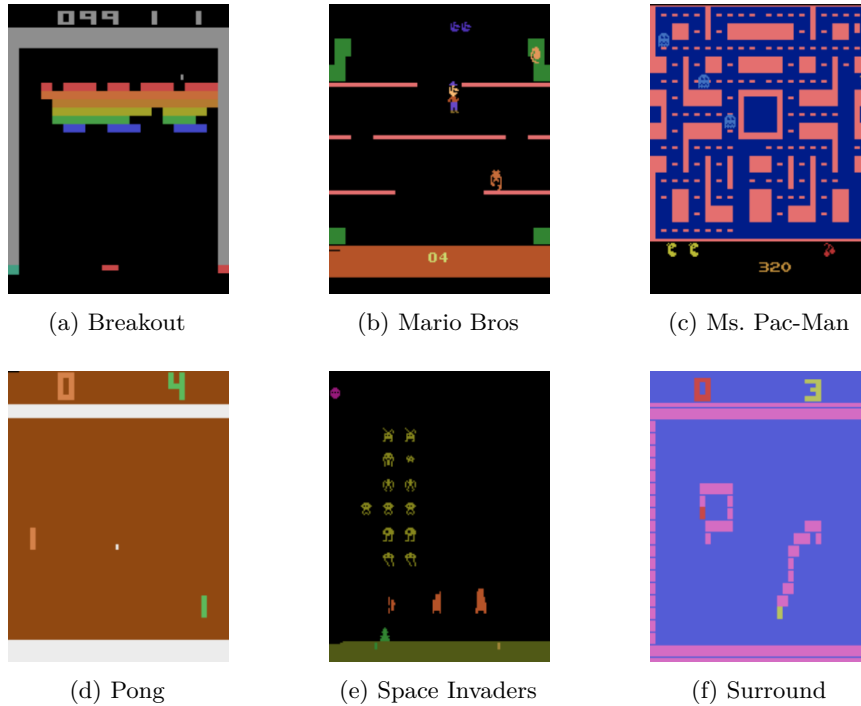


Figure 1.2: The six Atari environments used to evaluate CAR.

162 **Surround.** A two-player game on a grid (Figure 1.2f). Each player leaves a
 163 solid trail behind a moving cursor, and the goal is to force the opponent to crash
 164 into a trail or a wall.

165 1.2.2 Neural architecture

166 The initial policy and the surrogate policies are convolutional networks that share
 167 the same feature-extraction trunk. The trunk is a sequence of four convolutional
 168 layers, with 16, 32, 64, and 128 channels, 4×4 kernels, and a stride of 2, each
 169 followed by layer normalization and a LeakyReLU activation. Its input is the
 170 observation, a stack of four consecutive 64×64 frames.

171 The initial policy follows an actor-critic design: the trunk feeds two separate
 172 multilayer perceptrons, one producing the parameters of the action distribution
 173 and one producing the value estimate. The surrogate policy has the encoder-
 174 decoder structure introduced in Section ??: its encoder f_{enc} is a trunk with the
 175 same architecture, and its decoder is a single multilayer perceptron producing
 176 the parameters of the action distribution. The surrogate has no value head, since
 177 it is trained only to reproduce the action distribution of the initial policy.

178 The latent space analyzed by CAR is the representation produced by the
 179 last convolutional layer of the surrogate encoder.

Observations	
Frame size	64×64
Frames stacked per observation	4
Frame skip	5
Sticky-action probability	0.05
Color	grayscale (color for Ms. Pac-Man)
Initial policy training (PPO)	
Optimizer	Adam
Learning rate	1.5×10^{-4}
GAE λ	0.95
Policy clipping ε	0.1
Value-function clipping	10
Entropy coefficient	0.01
KL coefficient	0.5
Epochs per update	10
Gradient clipping (global norm)	100
Surrogate policy training (behavior cloning)	
Surrogate policies per environment	3
Observation–action pairs per dataset	$\geq 10^6$
Optimizer	Adam
Learning rate	1×10^{-4}
Batch size	2048
Loss	MSE between action-distribution parameters
Train / validation split	80 / 20
Latent space analysis	
Observations per surrogate policy	50 000
Clustering algorithm	k -means, Euclidean distance, 10 restarts
Number of clusters K	2 to 30
Points for the 2D projection	10 000

Table 1.2: Parameters common to all experiments, grouped by pipeline stage.

1.2.3 Parameters

Table 1.2 lists the parameters common to all experiments, grouped by pipeline stage. Parameters that vary across environments, such as the reward scaling and the exact number of training epochs, are given in the released configuration files.

1.3 Results

This section reports the results of applying CAR to the six environments described in Section 1.2. Sections 1.3.1 and 1.3.2 first check that the two learning stages of the pipeline behave as expected: that the initial policies reach a competent level of play, and that the surrogate policies faithfully imitate them. Section 1.3.3

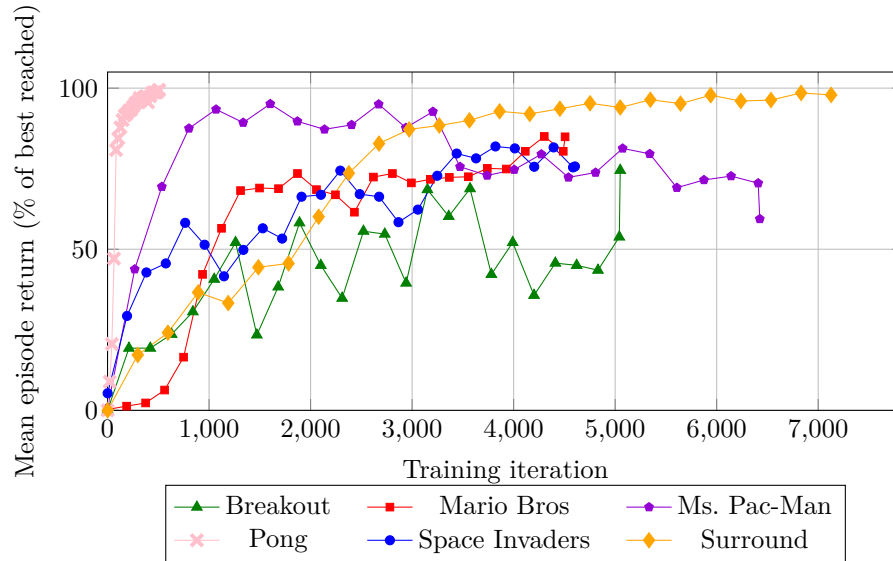


Figure 1.3: Training of the initial policies with PPO, one curve per environment.

189 then presents the analysis of the latent representations and the values of the
 190 three evaluation metrics.

1.3.1 Initial policy training

192 Figure 1.3 shows how the initial policies improve during PPO training. For
 193 each environment, the mean episode return is normalized so that 0% and 100%
 194 correspond to the lowest and highest mean return observed during that run. In
 195 every environment the return increases and then reaches a plateau, at which
 196 point training is stopped. The number of iterations required to reach this plateau
 197 varies from a few hundred, for Pong, to several thousand, for Surround and Ms.
 198 Pac-Man. For Ms. Pac-Man the return decreases after its peak; since the best
 199 checkpoint is retained rather than the last one, this does not affect the policy
 200 that is subsequently analyzed.

201 To confirm that the trained policies play competently, their scores are compared
 202 with the reference scores reported by [Machado et al., 2017]. Each initial
 203 policy reaches a score of the same order as these references, so it constitutes a
 204 reasonable target for imitation.

1.3.2 Surrogate policy training

206 Figure 1.4 shows, for each environment, the training action loss of the three
 207 surrogate policies: the mean squared error between the action-distribution
 208 parameters of the surrogate and those of the initial policy. In every case the loss
 209 decreases smoothly, and the three independently trained policies follow almost
 210 identical trajectories, converging to the same value; the three curves of each
 211 environment largely overlap. This consistency matters for CAR: the method
 212 compares the latent representations of independently trained policies, and this
 213 comparison is only meaningful if these policies implement the same decision

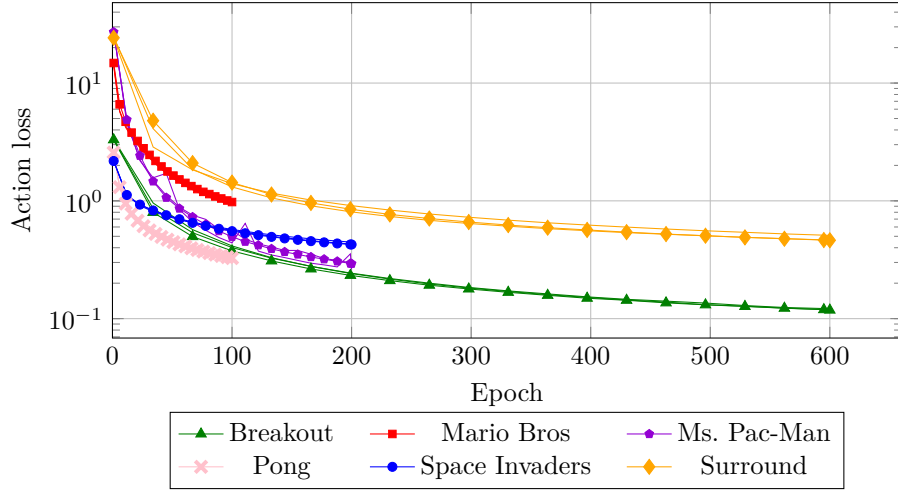


Figure 1.4: Training action loss of the surrogate policies, on a logarithmic scale, three curves per environment.

function. The behavioral fidelity of the surrogate policies is quantified by the *Behavior Reconstruction* in the next section.

1.3.3 Latent space analysis

We first evaluate the ability of the surrogate policies to replicate the behavior of the initial policies (*Behavior Reconstruction*, Table 1.3). The surrogate policies reach a *Behavior Reconstruction* of 0.96 on average over 100 evaluation episodes, ranging from 0.83 for Breakout to 1.00 for Ms. Pac-Man and Surround. They therefore reproduce the decisions of the initial policies closely enough for their latent representations to be analyzed in place of those of the initial policies.

Environment	<i>Behavior Reconstruction</i>	<i>Abstraction Score</i>	<i>Clustering Universality</i>
Breakout	0.83 $\pm$ 0.03	0.82 $\pm$ 0.07	0.48 $\pm$ 0.11
Mario Bros	0.99 $\pm$ 0.03	0.71 $\pm$ 0.14	0.55 $\pm$ 0.10
Ms. Pac-Man	1.00 $\pm$ 0.03	0.43 $\pm$ 0.09	0.65 $\pm$ 0.08
Pong	0.95 $\pm$ 0.01	0.99 $\pm$ 0.00	0.49 $\pm$ 0.05
Space Invaders	0.97 $\pm$ 0.07	0.69 $\pm$ 0.06	0.67 $\pm$ 0.07
Surround	1.00 $\pm$ 0.01	0.55 $\pm$ 0.07	0.61 $\pm$ 0.09

Table 1.3: Evaluation of *Behavior Reconstruction*, *Abstraction Score* and *Clustering Universality* across the six Atari environments.

For the remainder of the analysis, a random subset of 50,000 observations is embedded into the Latent Space (LS) of each surrogate policy, using the representation extracted directly after the last convolutional layer, and the resulting point cloud $\mathcal{E}$ is partitioned with k -means clustering [MacQueen, 1967].

Figure 1.5 reports the evolution of *Clustering Universality* with respect to the number of clusters. After the first few values of K , every curve settles onto a plateau whose level depends on the environment, from about 0.45–0.50 for Breakout and Pong to about 0.65 for Ms. Pac-Man and Space Invaders. Since *Clustering Universality* is computed from the Adjusted Rand Index (ARI), and an ARI of 0.5 already corresponds to substantial agreement [Amezquita et al., 2020], these values indicate that independently trained surrogate policies recover a similar cluster structure in every environment, and that this structure is stable with respect to the number of clusters. The strength of this agreement varies across environments, but remains clearly above chance everywhere. The absence of a single optimal number of clusters suggests that K should be chosen according to the level of granularity desired for the analysis.

The *Abstraction Score*, in contrast, varies markedly across environments. It is close to 1 for Pong (0.99) and high for Breakout, Mario Bros and Space Invaders (0.69 to 0.82): in these environments the latent clusters are largely independent of a direct clustering of the observations or of the action distributions. It is lower for Surround (0.55) and Ms. Pac-Man (0.43). In these two environments the position of the agent on the screen largely determines both the raw observation and the decision to take, so a clustering of the LS naturally coincides with a clustering of the observations, and the *Abstraction Score* cannot separate the two; a low score does not by itself imply that the representation is not abstract.

With fewer clusters, CAR highlights broad strategic patterns. For instance, in the Mario Bros environment, the two main groups correspond to whether the POW block is present (highlighted in red in Figure 1.7A) or not. This is particularly interesting, as the use of the POW block is a key determinant of strategy in this environment [AtariAge, 2024].

With a larger number of clusters, the analysis captures more fine-grained tactical behaviors, such as “Attacking the left” and “Navigating through the level” (see Figure 1.7B).

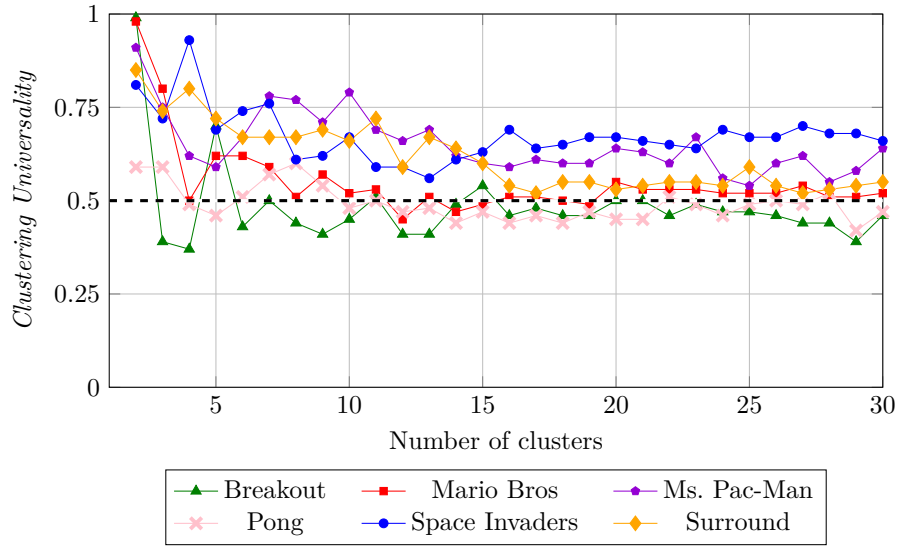


Figure 1.5: Evolution of *Clustering Universality* with respect to the number of clusters, for each of the six Atari environments.

Both perspectives are valid, depending on whether the goal is to study strategic or tactical behaviors. Similar interpretable structures can be observed across the evaluated environments.

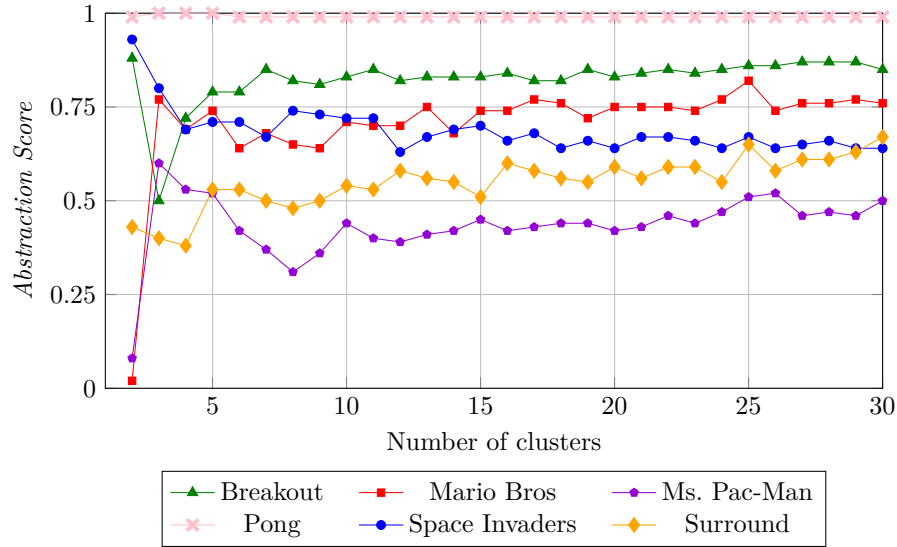
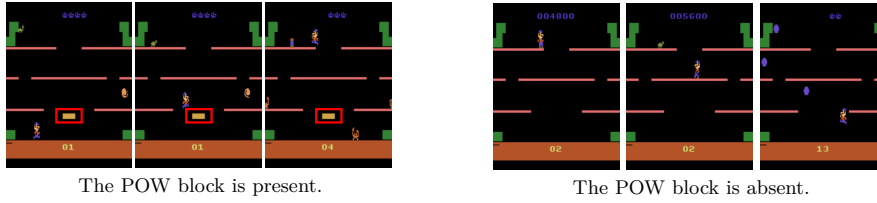


Figure 1.6: Evolution of *Abstraction Score* with respect to the number of clusters, for each of the six Atari environments.

(A) The case with 2 clusters.



(B) The case with 5 clusters.

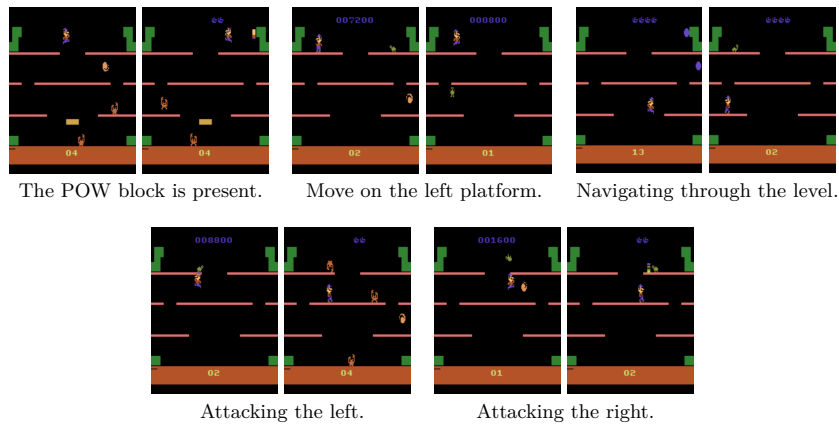


Figure 1.7: Effect of the number of clusters on the groupings identified by CAR in the Mario Bros environment.

Acronyms

APPO Asynchronous Proximal Policy Optimization 2

ARI Adjusted Rand Index 10

CAR Clustering Agent Representation 1–4, 6–8, 10, 12

DL Deep Learning 1, 2

DQN Deep Q-Networks 2

GPU Graphics Processing Unit 3

LS Latent Space 10

PPO Proximal Policy Optimization 2, 8

RL Reinforcement Learning 1–5

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